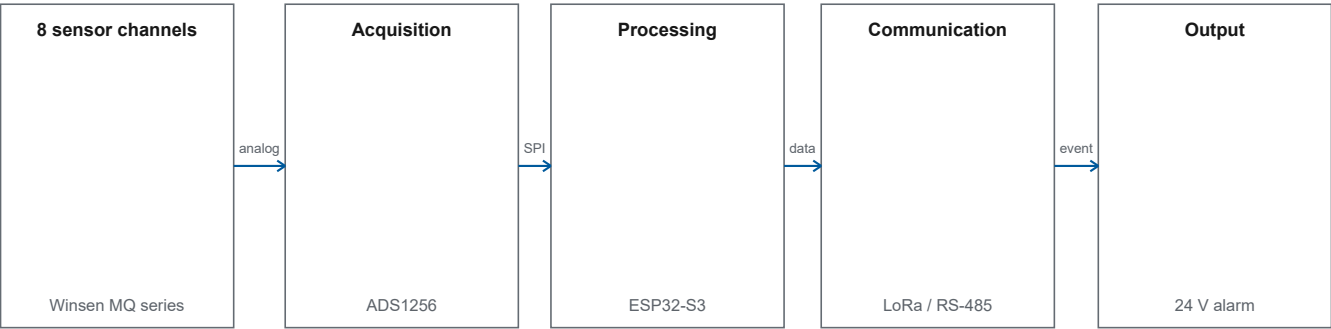


GasleakDetector

Technical Datasheet Revision 4.0

Eight-channel sensing device with local processing, 24 V alarm output, LoRa STAR, and RS-485.



SENSOR CHANNELS	MAIN SUPPLY	RF	ALARM
8 channels	24 VDC nominal	920-923 MHz	AUTO, 24 V output

Key features

- Acquires eight sensor channels with per-channel power control.
- Runs local processing in Running/Inference mode.
- Sends telemetry and alarm events to CH over LoRa STAR.
- Provides a steady 24 V output for an external alarm device.

Technical overview

GasleakDetector combines multi-channel sensing, local processing, a physical alarm output, and two communication interfaces in one field endpoint.

Functional profile

Architecture	8 analog channels
Control	ESP32-S3
Primary link	LoRa STAR to CH
Local link	RS-485 Modbus

Operational interfaces

Local processing	Scanning, input validation, inference, status, and frame construction.
Communication	LoRa STAR as the primary field path; RS-485 as a local read-only interface.
Alarm	AUTO follows a valid alarm inference; MANUAL is restricted to a test session.

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1 Product identification

Product	GasleakDetector
Product status	Engineering Prototype
Document revision	4.0
Technical baseline	Firmware 0.8.19 Protocol 0.2.0
Issuer	Lab IoT ITB

1.1 Abbreviations

Abbreviation	Meaning	Abbreviation	Meaning
ADC	Analog-to-Digital Converter	RTU	Remote Terminal Unit in Modbus
DAC	Digital-to-Analog Converter	STAR	Endpoint-to-CH radio domain
NVM	Non-volatile memory	TLS	Transport Layer Security
RF	Radio frequency		

2 Overview and architecture

2.1 Product overview

GasleakDetector combines multi-channel sensing, local processing, a physical alarm output, and two communication interfaces in one field endpoint.

Architecture	8 analog channels	Control	ESP32-S3
Primary link	LoRa STAR to CH	Local link	RS-485 Modbus

- Acquires eight sensor channels with per-channel power control.
- Runs local processing in Running/Inference mode.
- Sends telemetry and alarm events to CH over LoRa STAR.
- Provides a steady 24 V output for an external alarm device.

2.2 Functional architecture

The primary path runs from sensors through acquisition and processing to communication, alarm, or telemetry.

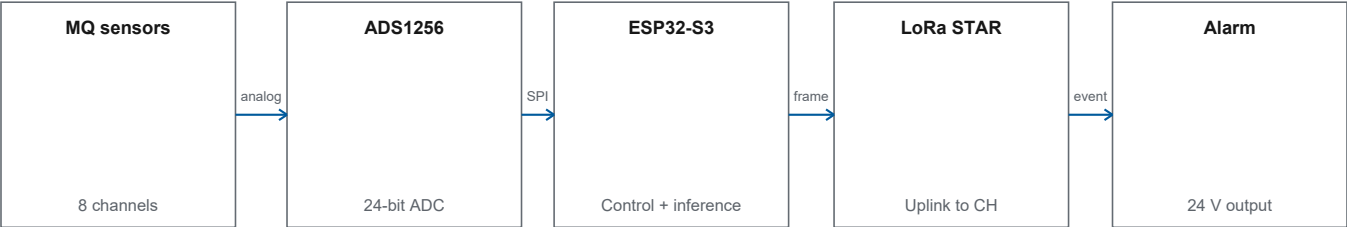


Figure 1. Primary GasleakDetector functional path.

Local processing	Scanning, input validation, inference, status, and frame construction.
Communication	LoRa STAR as the primary field path; RS-485 as a local read-only interface.
Alarm	AUTO follows a valid alarm inference; MANUAL is restricted to a test session.

2.3 Primary hardware components

This table identifies components present in the design; component ratings do not automatically become assembled-product ratings.

Function	Component	Product role
MCU	ESP32-S3-WROOM-1U-N16R8	Control, communication, and local processing
ADC	ADS1256IDBR	24-bit multi-channel analog acquisition
Environment	SHT40-AD1B-R2	Auxiliary temperature and humidity sensing
Radio	E22-900MM22S	LoRa STAR interface
RS-485	THVD1410DR	Modbus RTU physical layer
Expander	PCF8574T	Sensor-channel enable control

3 Sensor channels and signal path

3.1 Eight-channel sensor map

Channel order is part of the acquisition and processing contract; gas performance is not inferred from sensor names.

Channel	Sensor	Identification
1	MQ-8	Winsen MQ series
2	MQ-135	Winsen MQ series
3	MQ-3	Winsen MQ series
4	MQ-5	Winsen MQ series
5	MQ-4	Winsen MQ series
6	MQ-7	Winsen MQ series
7	MQ-6	Winsen MQ series
8	MQ-2	Winsen MQ series

3.2 Channel power control and nulling

Each channel has a control path that enables directed sensor activation and nulling.

Stage	Description
Control layer	ESP32-S3 controls enables and service sequence
I/O expansion	PCF8574T maps eight enable signals
Power switch	TPS22919 controls supply per channel
Analog nulling	DAC and multiplexer apply channel correction

Figure 2. Power-control and nulling layers.

- Nulling mode is an engineering function separate from normal operation.
- Channel status and nulling results are available for engineering readback.

Note: The nulling procedure covers all eight channels, profile storage, reboot, and re-application of a valid profile.

3.3 Analog acquisition path

Acquisition uses an external 24-bit ADC under sequenced firmware control.

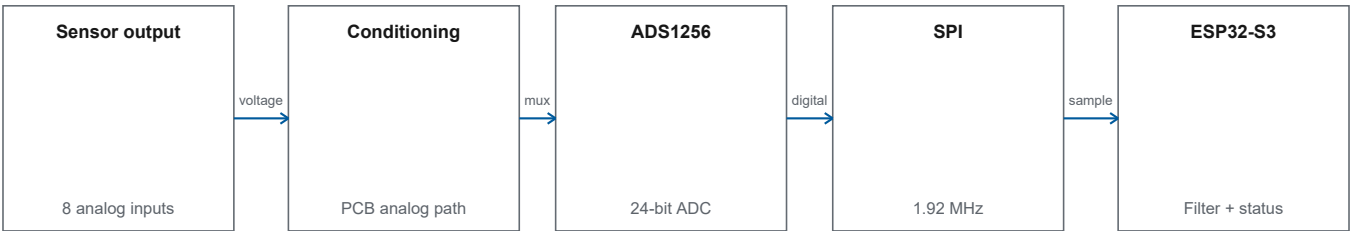


Figure 3. Implemented analog acquisition chain.

ADC	Texas Instruments ADS1256IDBR, 24-bit component resolution.
Firmware SPI clock	1,920,000 Hz
Processing output	Channel values, validity, fault status, and inference input.

3.4 ADC and sampling configuration

Firmware explicitly controls multiplexing, gain, calibration, and sample acceptance.

Parameter	Implemented value	Meaning
ADS1256 data rate	30,000 SPS	ADC component configuration
Firmware VREF	2.497 V	Source conversion value
PGA	1x - 64x adaptive	Gain selected according to input
Calibration	Self-calibration	Performed in the initialization path
After mux switch	First conversion discarded	Reduces carry-over after channel switching
Default scan cadence	500 ms	Firmware acquisition-cycle interval

3.5 Filtering and valid-batch behavior

Processing commits a batch only when all eight channels pass validity and saturation checks.

Stage	Description
Scan 8 channels	Multiplexed sequential acquisition
Per-channel checks	Read success, validity, saturation
Batch gate	Must be 8/8 valid and non-saturated
Moving average	10 samples per channel
Commit	Data becomes processing/inference input

Figure 4. Sensor-batch acceptance gates.

Note: One invalid channel makes the complete batch invalid.

3.6 Inference-validity gates

A classifier result is valid only when acquisition, model, nulling, and DAC state are all ready.

Stage	Description
Sensor gate	8/8 channels valid and non-saturated
Filter gate	Moving average is primed
Model gate	ML runtime and model metadata are ready
Nulling gate	Profile is applied and bound to the model
DAC gate	Nulling output is verified
Valid inference	Only then may affect AUTO alarm

Figure 5. Layered gates before inference is declared valid.

4 Inference and operating modes

4.1 Configured classification outputs

The inference model provides four classification labels and a confidence value.

Class 0	Clean Air	Class 1	LPG
Class 2	H2	Class 3	CO2

Output	Meaning
Class label	Configured model classification result
Confidence	Classifier confidence value

4.2 Auxiliary temperature and humidity sensing

SHT40 provides auxiliary temperature and humidity data for status and processing.

Component	SHT40-AD1B-R2	Interface	I2C
Quantities	Temperature + RH	Role	Auxiliary input

- Firmware converts raw digital temperature and humidity values.
- Values can be included in status/telemetry according to the firmware path.

Note: Official component reference: Sensirion SHT40 product page and SHT4x datasheet.

4.3 Firmware operating modes

Modes are separated by operational, data-collection, and commissioning purpose.

Mode	Purpose	Use boundary
Running / Inference	Scanning and local inference	Normal operating mode after boot
Dataset	Engineering data collection	Not a production-alarm path
Nulling	Per-channel baseline correction	Requires engineering workflow
Battery session	Operating cycle when battery source is selected	Power behavior depends on hardware
MANUAL alarm	Alarm-load test	Session-only; AUTO returns at boot

Note: AUTO is the product default at every boot. MANUAL is not stored as a permanent boot selection.

4.4 Boot and normal-session sequence

This sequence shows the primary dependencies before the device can produce operational data.

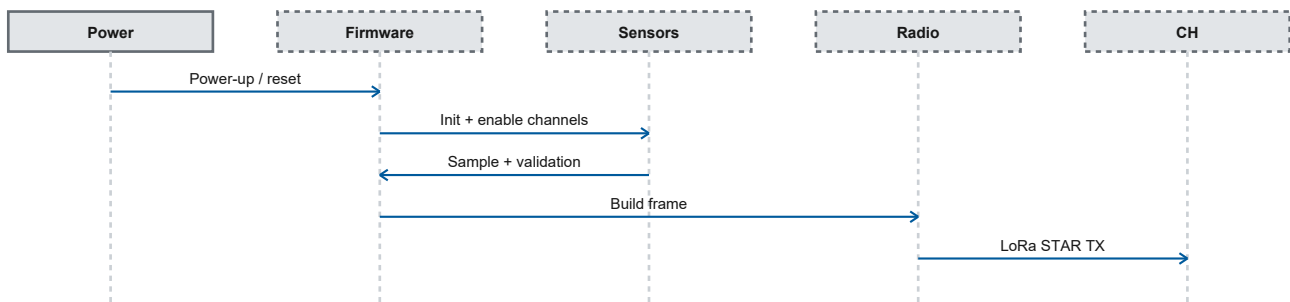


Figure 6. Boot-to-telemetry sequence.

- Initialization or validation failures are exposed through diagnostic status.
- Runtime configuration may come from firmware defaults or validated NVM.
- Actual RF configuration must be checked through device readback.

5 Alarm and power

5.1 Local alarm architecture

Firmware controls one 24 V alarm output; audible/visual cadence is generated by the external alarm device.

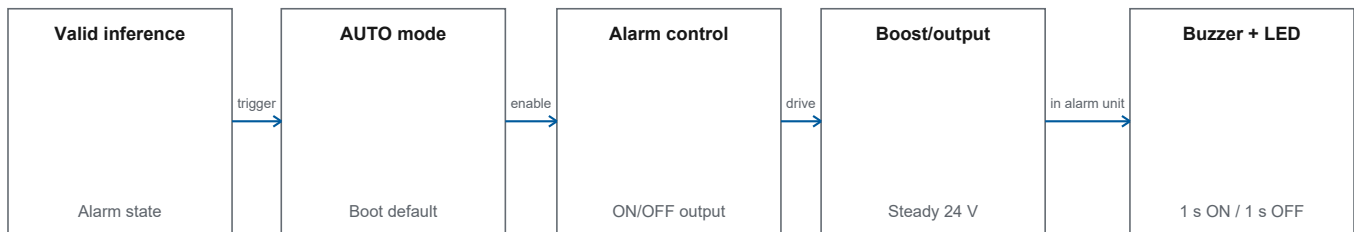


Figure 7. Firmware and external-alarm responsibility boundary.

Firmware behavior	Steady 24 V ON/OFF output.
Load behavior	Buzzer and LED automatically cycle 1 second ON, 1 second OFF from steady 24 V.
Test mode	Session-only MANUAL mode selectable through Operator Hub.

Note: The 24 V output has been tested with an external buzzer-and-LED alarm device.

5.2 Power architecture

The product accepts 24 VDC as its main supply and includes a separate battery operating path.

Stage	Description
Main input	24 VDC nominal
Source selection	Battery is selected when detected; otherwise 24 V/5 V follows hardware status
Internal rails	Conversion for MCU, analog, radio, and sensors
Alarm output	Steady 24 V when active

Figure 8. Primary product power domains.

Electrical parameter	Value
Nominal product input	24 VDC
Measured current	Up to 300 mA steady with eight sensors active

5.3 Current consumption at the 24 V input

Values are steady-state measurements as the number of active sensors increases from zero to eight.

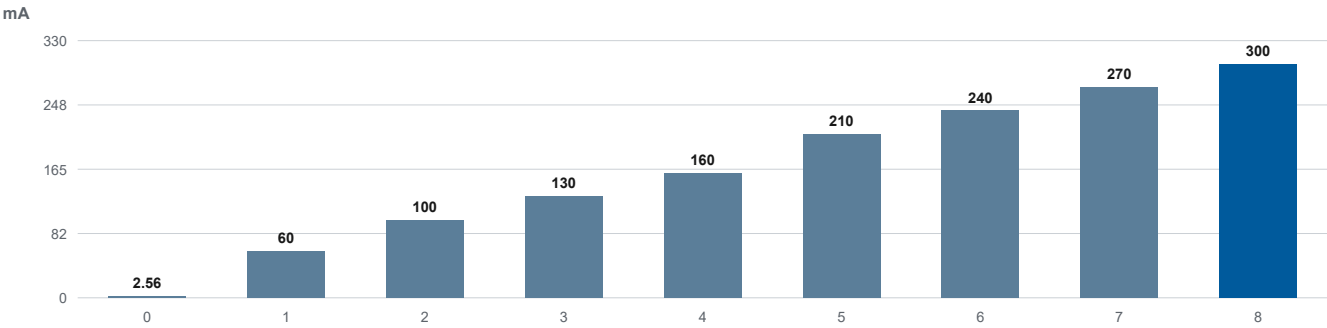


Figure 9. Steady-state current versus active sensor count.

Active sensors	0	1	2	3	4	5	6	7	8
Current (mA)	2.56	60	100	130	160	210	240	270	300

Note: The meter was placed in series in the 24 V+ path from PSU to board. The 300 mA value is steady-state, not peak current or PSU rating.

5.4 Battery configuration

Battery mode uses seven 18650 cells connected in parallel.

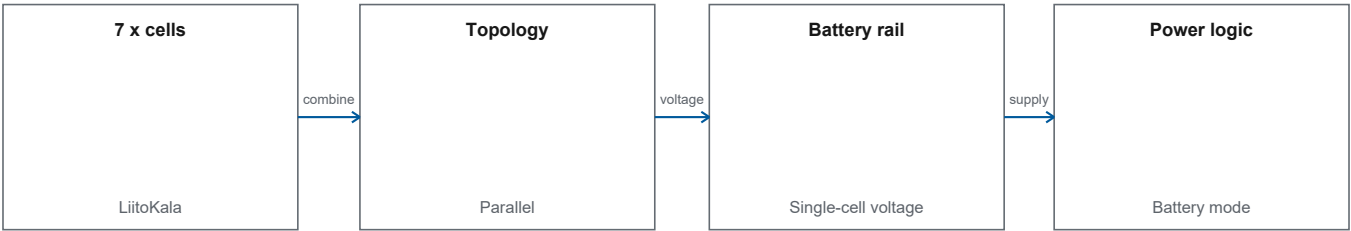


Figure 10. GasleakDetector battery topology.

Cell type	LiitoKala Lii-King4000 18650, 4000 mAh label per cell
Quantity	7
Arrangement	Parallel

Note: The 4000 mAh figure is a cell-label value, not a product-pack capacity-test result.

5.5 Battery-voltage monitoring

Firmware reports battery voltage and diagnostic flags but does not use those flags as an active cutoff.

Diagnostic	Source threshold	Behavior
Battery low	3.50 V	Status/flag; does not stop operation
Battery critical	3.30 V	Status/flag; not a physical cutoff

Stage	Description
Voltage read	Battery-monitor acquisition
Diagnostic compare	Low / critical flags
Status output	Value and flags available for readback
Protection boundary	Does not actively disconnect pack or rail

Figure 11. Diagnostic monitoring versus active protection.

6 Communication interfaces

6.1 LoRa STAR interface

LoRa STAR connects GasleakDetector to the STAR radio on CH.

Parameter	Firmware default	Note
Carrier	920.0 MHz	Configuration input constrained to 920-923 MHz inclusive
Bandwidth	125 kHz	Must match GasleakDetector and CH STAR
Spreading factor	SF7	STAR-domain default
Coding rate	4/5	Firmware configuration value CR=5
TX power	17 dBm	Firmware setting, not measured output
Preamble	8	Firmware default
Antenna	3 dBi	Product configuration

Note: The active configuration is verified through device readback after commissioning.

6.2 RS-485 / Modbus interface

The local interface is provided as a read-only Modbus RTU slave for product-exposed data/status.

Physical layer	RS-485	Protocol	Modbus RTU
Mode	Read-only slave	Unit ID	1
Serial format	9600 bit/s, 8N1		
Transceiver	THVD1410DR		
Customer access	Reads the eight supported public registers; not used to write product control.		

Note: The interface uses function codes 03/04 and the read-only registers defined in the product register map.

6.3 Data and frame security

Field payloads use application authentication so the receiver can verify frame authenticity and reject invalid frames before further processing.

Stage	Description
Sensor data	Values, status, and events
Application frame	Identity, sequence, and payload
AES-128-GCM	Confidentiality + authentication
LoRa transport	STAR toward CH

Figure 12. Field-payload protection layers.

- CH forwards field frames and is not a TLS termination point.
- TLS, when selected, is on the Gateway-to-broker segment.
- The receiver rejects payloads that fail authentication or anti-replay checks.

7 Commissioning and diagnostics

7.1 Commissioning through Operator Hub

Operator Hub is the engineering interface for firmware-variant selection, upload, configuration, and readback.

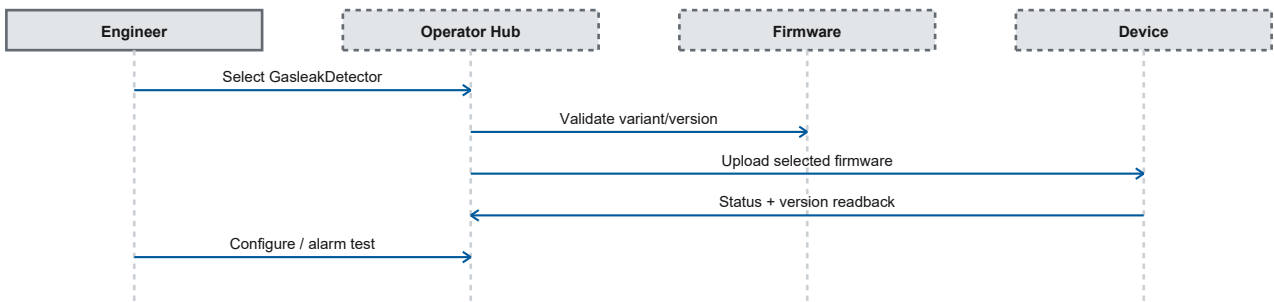


Figure 13. Operator-controlled commissioning flow.

Commissioning verification	Criterion
Version and hardware	The active version and hardware profile are verified through device readback after upload.
Alarm mode	MANUAL mode requires an explicit step and returns to AUTO on boot.
RF boundary	Frequency configuration outside 920-923 MHz is rejected.

7.2 Status and diagnostics

Machine-readable status separates configuration, subsystem readiness, faults, and operational results.

Status group	Engineering content examples
Identity	Firmware version, protocol version, device identity
Sensors	Active channels, validity, and fault state
Nulling	Profile/status and per-channel result
Radio	Frequency, bandwidth, SF, CR, power, preamble
Alarm	AUTO/MANUAL, commanded output, session-only flag
Power	Detected source and power-path state

Note: Device readback verifies the active unit configuration.

8 Operational parameters and interfaces

8.1 Operational timing

The firmware runs acquisition, radio reporting, and the receive window at the following fixed intervals.

Function	Value	Description
Sensor scan cycle	500 ms	Processing interval for the eight-channel measurement.
Periodic radio report	10 s	Nominal interval for operational data transmission over LoRa STAR.
Post-transmission receive window	2 s	Time during which the device keeps the receive path available after transmission.
Sensor warm-up in battery session	30 s	The eight sensor channels warm up before the battery-session inference cycle.

8.2 Modbus RTU interface

RS-485 provides eight read-only 16-bit registers for integration with an external controller or acquisition system.

Parameter	Specification
Mode	Modbus RTU / RS-485
Unit address	1
Serial format	9600 bit/s, 8 data bits, no parity, 1 stop bit (8N1)
Read functions	03 (Holding Registers), 04 (Input Registers)
Access	Read-only

Address	Name	Content
0	Status	Device status bits; see the status table.
1	Gas class	Inference-result class identifier.
2	Confidence	Confidence value in percent.
3	Battery voltage	Battery reading in millivolts.
4	24 V input source	Indicates that the selected input source is 24 V.
5	External power	External-power-active flag.
6	Transmission counter	Low 16 bits of the LoRa transmission counter.
7	Node ID	Device node identifier.

8.3 Device status definition

The status register uses independent bits so that an external system can distinguish device readiness, inference validity, power sources, and alarm state.

Bit	Meaning when set
0	ADC acquisition ready
1	DAC ready
2	Radio ready
3	Inference model ready
4	Inference result valid
5	Alarm condition active
6	Selected input source is 24 V
7	External power active

8.4 Alarm-output logic

The 24 V alarm output provides AUTO mode for normal operation and MANUAL mode for controlled testing.

Mode	Control condition	24 V alarm output
AUTO	Inference asserts alarm	Active
AUTO	Inference does not assert alarm; clear-state persistence succeeds	Inactive
AUTO fail-safe	Clear-state persistence fails	Remains active until a valid clear can be persisted
MANUAL	Test command ON	Active
MANUAL	Test command OFF	Inactive

Note: MANUAL mode is not stored as a permanent operating mode. After a device restart, the initial mode is AUTO. Failure to persist the clear state retains the alarm ON as a fail-safe. When the 24 V output is active, the one-second ON/one-second OFF audible and visual pattern is generated by the powered alarm device.

8.5 Battery monitoring

Battery voltage is filtered for telemetry and diagnostics. The following thresholds are monitoring thresholds, not automatic power-disconnection functions.

Parameter	Value	Function
Samples per reading	16	ADC reading averaging.
Voltage-divider ratio	3.0	Conversion from ADC voltage to battery voltage.
Valid-reading threshold	3.0 V minimum	Rejects readings below the validity threshold.
Low threshold	3.5 V	Low-battery monitoring state.
Critical threshold	3.3 V	Critical-battery monitoring state.
Filter coefficient	0.20	Exponential moving average.

Note: The firmware monitors and reports battery voltage but does not prevent boot, stop transmission, or enter a low-power mode based on these thresholds.

9 References and revision history

External references describe components; assembled-product values follow the specifications in this document.

Reference	Document / URL
R1	Espressif - ESP32-S3-WROOM-1/1U Datasheet, v1.8, documentation.espressif.com
R2	Texas Instruments - ADS1256 product datasheet, ti.com/product/ADS1256
R3	Texas Instruments - THVD1410 product datasheet, ti.com/product/THVD1410
R4	Sensirion - SHT40 / SHT4x Datasheet, sensirion.com/products/catalog/SHT40
R5	Winsen - MQ Sensor Series Overview, winsen-sensor.com/mq-sensor.html

Revision	Change
3.0	Updated technical structure and specifications.